

Rizuan

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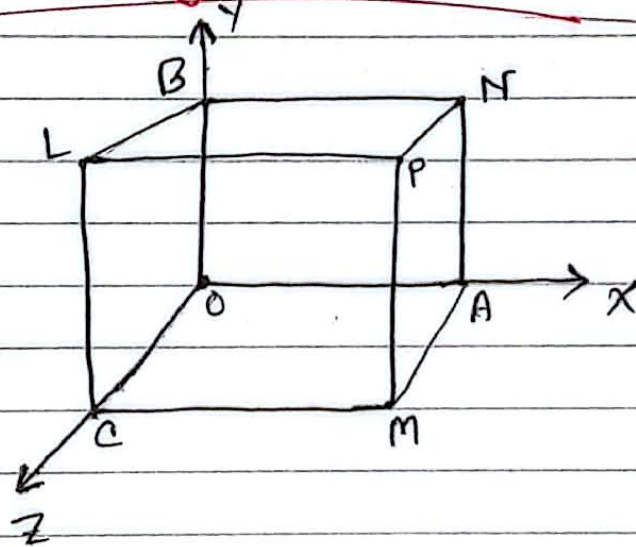
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Rectangular Co-ordinate.



The position of a point in space may be defined to these three fixed planes. These planes have a common point known as origin, intersect in pairs in three straight lines known as co-ordinates axes, planes are called co-ordinate planes. When the three co-ordinate planes are naturally at right angles, the axes are said to be rectangular. The three planes divided the spaces into 6 parts known as hexons.

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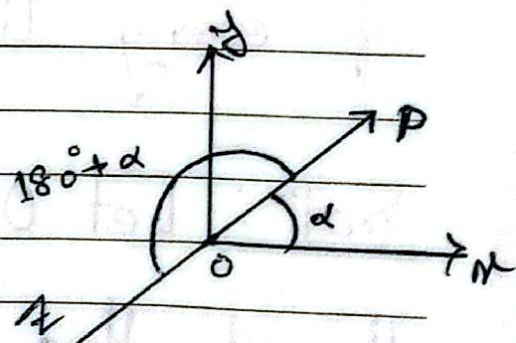
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Let p be the point in space, the three numbers x, y, z are called the co-ordinates of the point P and is represented as (x, y, z)

Direction of cosines of a line : —

If a given line OP makes angles α, β, γ with the positive direction of axes of x, y, z respectively, then $\cos \alpha, \cos \beta, \cos \gamma$ are the direction cosines of the line OP .

The angles, α, β, γ of the line OP are called the direction angles of OP .



$$\cos \alpha = l, \cos \beta = m, \cos \gamma = n$$

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Direction Ratios:

Any three numbers a, b and c which is proportional to the direction cosines l, m, n respectively of a line are called direction ratios of the given line -

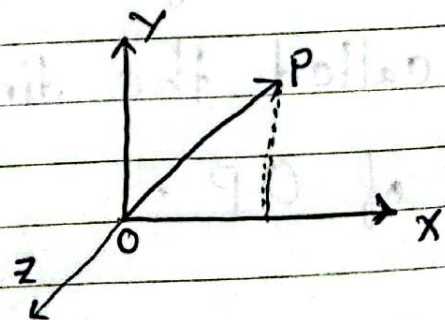
$$\textcircled{*} \frac{l}{a} = \frac{m}{b} = \frac{n}{c} = \pm \frac{\sqrt{l^2 + m^2 + n^2}}{\sqrt{a^2 + b^2 + c^2}}$$

Relation between direction cosines:

Or, prove that, $l^2 + m^2 + n^2 = 1$

Or, the sum of the squares of the direction of every line is one.

Solⁿ: Let OP be drawn



through the origin parallel to the given lines. so that, l, m, n are

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the cosines of angles which makes with OX , OY and OZ respectively.

Let (x, y, z) be the co-ordinates of any point P on the line.

Let, $OP = r$

$$\therefore \cos \alpha = x/r ; \cos \beta = y/r ; \cos \gamma = z/r$$

$$\therefore l = x/r ; m = y/r ; n = z/r$$

$$\therefore x = lr ; y = mr ; z = nr$$

— (i)

Squaring and adding we get,

$$x^2 + y^2 + z^2 = (l^2 + m^2 + n^2) r^2$$

$$\Rightarrow r^2 = (l^2 + m^2 + n^2) r^2$$

$$\Rightarrow l^2 + m^2 + n^2 = 1 \quad \text{(A.B.)}$$

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Find the direction cosines of the line drawn from origin to the point $(-6, 2, 3)$.

Soln:

$$\cos \alpha = \frac{x_2 - x_1}{\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}}$$

$$= \frac{-6 - 0}{\sqrt{(-6 - 0)^2 + (2 - 0)^2 + (3 - 0)^2}}$$

$$= \frac{-6}{\sqrt{49}}$$

$$= -6/7.$$

$$\cos \beta = \frac{y_2 - y_1}{\sqrt{49}} = \frac{2 - 0}{7} = \frac{2}{7}$$

$$\cos \gamma = \frac{z_2 - z_1}{\sqrt{49}} = \frac{3 - 0}{7} = \frac{3}{7}$$

Ans

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Find the acute angle between the lines whose direction cosines are given by the relation $l+m+n=0$, and $l^2+m^2+n^2=0$:

Soln: We have,

$$l+m+n=0 \quad \text{--- (i)}$$

$$l^2+m^2+n^2=0 \quad \text{--- (ii)}$$

From eqn (i), $n = -(l+m)$

putting the value in eqn (ii)

$$l^2+m^2+n^2=0$$

$$\Rightarrow l^2+m^2 - (l+m)^2 = 0$$

$$\Rightarrow l^2+m^2 - l^2 - 2lm - m^2 = 0$$

$$\Rightarrow -2lm = 0$$

$$\Rightarrow l=0 \quad \text{or} \quad m=0 \quad \text{--- (4)}$$

--- 3)

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From, eqn i) & 3), where,

$$l + m + n = 0$$

$$\therefore 1 \cdot 1 + 0 \cdot m + 0 \cdot n = 0$$

By cross product,

$$\frac{l}{0-0} = \frac{m}{1-0} = \frac{n}{0-1} = \frac{\sqrt{l^2+m^2+n^2}}{\sqrt{0^2+1^2+(-1)^2}} = \frac{1}{\sqrt{2}}$$

$$\text{From i) \& 4) } \therefore l_1 = 0, m_1 = \frac{1}{\sqrt{2}}, n_1 = \frac{-1}{\sqrt{2}}$$

$$l + m + n = 0$$

$$0 \cdot l + 1 \cdot m + 0 \cdot n = 0$$

$$\therefore \frac{l}{0-1} = \frac{m}{0-0} = \frac{n}{1-0} = \frac{\sqrt{l^2+m^2+n^2}}{\sqrt{(1)^2+0^2+1^2}} = \frac{1}{\sqrt{2}}$$

$$\therefore l_2 = \frac{-1}{\sqrt{2}}, m_2 = 0, n_2 = \frac{1}{\sqrt{2}} \quad \text{(A)}$$

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Ex: Direction cosine of two straight lines are connected by the relation $2l + 2m - n = 0$ and $mn + nl + lm = 0$, show that the lines are perpendicular each other.

Solⁿ: We have,

$$2l + 2m - n = 0 \quad \text{--- 1)}$$

$$mn + nl + lm = 0 \quad \text{--- 2)}$$

From 1), $n = 2l + 2m$

putting value in 2)

$$m(2l + 2m) + (2l + 2m)l + lm = 0$$

$$\Rightarrow 2lm + 2m^2 + 2l^2 + 2ml + lm = 0$$

$$\Rightarrow 2m^2 + 5lm + 2l^2 = 0$$

$$(2m + l)(2l + m) = 0$$

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From eqn. 1) & 3)

$$2l + 2m - n = 0$$

$$2m + l = 0$$

$$\Rightarrow \frac{l}{0+1} = \frac{m}{-2-0} = \frac{n}{2-4}$$

$$\Rightarrow \frac{l}{1} = \frac{m}{-2} = \frac{2n}{-2} = \frac{\sqrt{l^2+m^2+n^2}}{\sqrt{1^2+(-2)^2+(-4)^2}} = \frac{1}{3}$$

$$\therefore l_1 = \frac{1}{3}, m_1 = \frac{-2}{3}, n_1 = \frac{-2}{3}$$

$$2l + 2m - n = 0$$

$$2l + m = 0$$

$$\therefore \frac{l}{0+2} = \frac{m}{-1-0} = \frac{n}{4-2}$$

$$\Rightarrow \frac{l}{2} = \frac{m}{-1} = \frac{n}{4-2} = \frac{\sqrt{l^2+m^2+n^2}}{\sqrt{4+1+4}} = \frac{1}{3}$$

$$\therefore l_2 = \frac{2}{3}, m_2 = \frac{-1}{3}, n_2 = \frac{2}{3}$$

Thus, the lines are perpendicular.

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Plane :

Def: A plane is a surface such that if any two points are taken on it, the straight line joining them lies wholly in the surface.

Corolla: General equation of a given point (x_1, y_1, z_1)

$$a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$$

C-2: The equation of a plane passing through two points,

(x_1, y_1, z_1) & (x_2, y_2, z_2)

$$(x - x_1)(y - y_1) - (x - x_2)(y - y_1)$$

$$= A \{ (x - x_1)(z - z_1) - (x - x_2)(z - z_1) \}$$

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Q-3: plane passing through three points:

$$\begin{vmatrix} x & y & z & 1 \\ x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \end{vmatrix} = 0$$

Ex-1: Find the equation of the plane passing through the intersection of the planes $x + 2y + 3z + 4 = 0$ and $4x + 3y + 2z + 1 = 0$ and the point $(1, 2, 3)$.

Solⁿ: Any plane through the intersection of two plane is -

$$(x + 2y + 3z + 4) + k(4x + 3y + 2z + 1) = 0 \quad \text{--- (i)}$$

since it passes through $(1, 2, 3)$

$$18 + 17k = 0 \Rightarrow k = -\frac{18}{17}$$

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putting the value in eqn i)

$$(x+2y+3z+4) + \left(-\frac{18}{17}\right) (4x+3y+2z+1) = 0$$

$$\Rightarrow 11x + 4y - 3z = 10. \quad \underline{\underline{(A)}}$$

Ex-2' The eq. equation of the plane through

the point $(4, -2, 1)$ and parallel to the

plane whose direction ratios are $7, 2, -3$,

Soln: The equation of the plane the point $(4, -2, 1)$

$$a(x-x_1) + b(y-y_1) + c(z-z_1) = 0$$

$$\Rightarrow a(x-4) + b(y+2) + c(z-1) = 0$$

It is parallel to the point plane whose

direction ratio's are $a=7, b=2, c=-3$

$$7(x-4) + 2(y+2) - 3(z-1) = 0$$

$$\Rightarrow 7x + 2y - 3z = 21.$$

(A)

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Ex-3: Find the equation of the plane through the point $(4, 0, 1)$ and parallel to the plane $4x + 3y - 12z + 6 = 0$

Solⁿ: The required plane,

$$4x + 3y - 12z + k = 0$$

It passes through the point $(4, 0, 1)$

$$4 \cdot 4 + 3 \cdot 0 - 12 \cdot 1 + k = 0$$

$$\Rightarrow k = -4.$$

The required plane is $4x + 3y - 12z - 4 = 0$

Ex-4: Find the equation of the plane passing through the lines of the intersection of the planes $2x - y = 0$ and $3z - y = 0$ and perpendicular to the plane $4x + 5y - 3z + 1 = 0$

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Soln:

The required plane,

$$(2x - y) + k(3z - y) = 0$$

$$\Rightarrow 2x - (1+k)y + 3kz = 0.$$

It is perpendicular to $4x + 5y - 3z + 1 = 0$

$$\Rightarrow 2 \cdot 4 - (1+k) \cdot 5 + 3k(-3) = 0$$

$$\Rightarrow k = -3/14.$$

The required plane is:

$$2x - y + 3/14(3z - y) = 0$$

$$\Rightarrow 28x - 17y + 9z = 0.$$

Ex-5: Find the equation of the plane through the points $(2, 2, 1)$ and $(9, 3, 6)$ and perpendicular to the plane $2x + 6y + 6z = 9$.

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Soln: The required plane is

$$\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ x_2-x_1 & y_2-y_1 & z_2-z_1 \\ a & b & c \end{vmatrix} = 0.$$

$$\Rightarrow \begin{vmatrix} x-2 & y-2 & z-1 \\ 9-2 & 3-2 & 6-1 \\ 2 & 6 & 6 \end{vmatrix} = 0.$$

$$\Rightarrow \begin{vmatrix} x-2 & y-2 & z-1 \\ 7 & 1 & 5 \\ 2 & 6 & 6 \end{vmatrix} = 0.$$

$$\Rightarrow (x-2) \begin{vmatrix} 1 & 5 \\ 6 & 6 \end{vmatrix} - (y-2) \begin{vmatrix} 7 & 5 \\ 2 & 6 \end{vmatrix} + (z-1) \begin{vmatrix} 7 & 1 \\ 2 & 6 \end{vmatrix} = 0$$

$$\Rightarrow 3x + 4y + 5z = 9.$$

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Ex-6: Find the eqn. of the plane, through

the line $\frac{x-2}{3} = \frac{y-3}{5} = \frac{z}{7}$ and passing

through the point $(1, -2, 3)$.

Soln: $\frac{x-2}{3} = \frac{y-3}{5}$

$$\Rightarrow 5x - 10 = 3y - 9$$

$$\Rightarrow 5x - 3y - 1 = 0$$

again, $\frac{x-2}{3} = \frac{z}{7}$

$$\Rightarrow 7x - 14 = 3z$$

$$\Rightarrow 7x - 3z - 14 = 0.$$

The req. plane,

$$(5x - 3y - 1) + k(7x - 3z - 14) = 0.$$

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It passes through $(1, -2, 3)$

$$(5 + 6 - 1) + k(-16) = 0$$

$$\Rightarrow k = \frac{10}{16} = \frac{5}{8}$$

from i),

$$(5x - 3y - 1) + \frac{5}{8}(7x - 3z - 14) = 0$$

$$\Rightarrow 25x - 8y - 5z - 26 = 0$$

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Transpose: —

$$A = \begin{bmatrix} 2 & 0 & 3 \\ 4 & 1 & 6 \\ 5 & 8 & 2 \end{bmatrix}$$

$$A_{11} = \begin{vmatrix} 1 & 6 \\ 8 & 2 \end{vmatrix} = 2 - 48 = -46.$$

$$A_{12} = \begin{vmatrix} 4 & 6 \\ 5 & 2 \end{vmatrix} = 8 - 30 = -22$$

$$A_{13} = \begin{vmatrix} 4 & 1 \\ 5 & 8 \end{vmatrix} = 32 - 5 = 27.$$

$$A_{21} = \begin{vmatrix} 0 & 3 \\ 8 & 2 \end{vmatrix} = 0 - 24 = -24$$

$$A_{22} = \begin{vmatrix} 2 & 3 \\ 5 & 2 \end{vmatrix} = 4 - 15 = -11$$

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$$A_{23} = \begin{vmatrix} 2 & 0 \\ 5 & 8 \end{vmatrix} = 16 - 0 = 16.$$

$$A_{31} = \begin{vmatrix} 0 & 3 \\ 1 & 6 \end{vmatrix} = 0 - 3 = -3$$

$$A_{32} = \begin{vmatrix} 2 & 3 \\ 4 & 6 \end{vmatrix} = 12 - 12 = 0$$

$$A_{33} = \begin{vmatrix} 2 & 0 \\ 4 & 1 \end{vmatrix} = 2 - 0 = 2$$

Now,

coefficient mat of A is

$$A = \begin{bmatrix} -4 & 6 & -2 & 2 & 27 \\ -2 & 4 & -1 & 1 & 16 \\ -3 & 0 & 2 & 1 & 2 \end{bmatrix}$$

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$$\det(A) = \begin{vmatrix} 1 & 6 & 4 & 6 & 4 & 1 \\ 2 & & -0 & & +3 & \\ 8 & 2 & & 5 & 2 & 5 & 8 \end{vmatrix}$$

$$= 2(2 - 48) - 0 + 3(32 - 5)$$

$$= -11$$

co-factor of A,

$$A = \begin{bmatrix} -4 & 6 & 22 & 27 \\ 2 & 4 & -11 & -16 \\ -3 & 0 & 2 & \end{bmatrix}$$

$$\therefore \text{Adj}(A) = \begin{bmatrix} -46 & 24 & -3 \\ 22 & -11 & 0 \\ 27 & -16 & 2 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \text{Adj}(A)$$

$$= \frac{1}{-11} \begin{bmatrix} -46 & 24 & -3 \\ 22 & -11 & 0 \\ 27 & -16 & 2 \end{bmatrix}$$

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Given,

$$A = \begin{bmatrix} 1 & -3 & -3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{bmatrix}$$

Characteristics of polynomial of A is,

$$\lambda I - A = 0$$

$$\Rightarrow \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & +3 & -3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{bmatrix} = 0$$

$$\Rightarrow \begin{bmatrix} \lambda - 1 & +3 & -3 \\ -3 & \lambda + 5 & -3 \\ 6 & -6 & \lambda - 4 \end{bmatrix} = 0$$

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$$\Rightarrow \begin{array}{c|cc|cc|cc} 1-1 & 1+5 & -3 & & -3 & -3 \\ & 6 & 1-4 & -(+3) & -6 & 1-4 \end{array}$$

$$\begin{array}{c|cc|cc} +(-3) & -3 & 1+5 & & & \\ & -6 & 6 & = & 0 & \end{array}$$

$$\Rightarrow (1-1) \{ (1+5)(1-4) + 18 \} - 3(-3+12-18)$$

$$-3(-18+6+30) = 0$$

$$\Rightarrow (1-1)(1^2-4+5-20+18) + (9+18)$$

$$-3(6+12) = 0$$

$$\Rightarrow (1-1)(1^2+1-2) + (9+18) - 18-36 = 0$$

$$\Rightarrow 1^3 + \cancel{1^2} - 2\cancel{1} - \cancel{1^2} - \cancel{1} + 2 + 9\cancel{1} + 18 - 18\cancel{1} - 36 = 0$$

$$\Rightarrow 1^3 - 12\cancel{1} - 16 = 0 \Rightarrow 1^3 - 4\cancel{1^2} + 4\cancel{1^2} + 16\cancel{1}$$

$$\Rightarrow 1^2(1-4) + 4\cancel{1}(1-4) + 16(1-4) = 0$$

$$\therefore 1 = 4, -2, 2$$